

A small commercial office (500–1,500 m²) replaces an ageing gas boiler with the hydronic split ASHP: the outdoor unit sits on a roof or external wall, refrigerant circuit contained entirely outside; the indoor hydraulic module connects to existing low-temperature radiators or new fan-coil units; a buffer vessel and controls interface with BMS for demand-based operation, achieving seasonal COP >3.2 and full MCS-compliant commissioning.

Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed
[D] Algorithmic Estimate

TARGET COST CEILING

£4,500.00
[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed
[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed
[B] Research

SUBSYSTEM MODULES

9

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source
- [D] Algorithmic estimate or LLM extraction
- [E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	PASS	Standards found	7 standards
3. Cost Ceiling Specified	PASS	Target provided	£4,500.00
4. Spatial Envelope Provided	PASS	Volume established	75 m³
5. Modules Decomposed	PASS	System architecture built	9 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	PASS	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements ^[A]

1.1 Mission & Use Case

Mission

To deliver a compact, R290-powered 30kW hydronic split heat pump that enables small UK commercial buildings to decarbonise heating cost-effectively and compliantly.

Use Case

A small commercial office (500–1,500 m²) replaces an ageing gas boiler with the hydronic split ASHP: the outdoor unit sits on a roof or external wall, refrigerant circuit contained entirely outside; the indoor hydraulic module connects to existing low-temperature radiators or new fan-coil units; a buffer vessel and controls interface with BMS for demand-based operation, achieving seasonal COP >3.2 and full MCS-compliant commissioning.

1.2 Market Context & Timing ^[B]

The UK Future Buildings Standard (effective 2025), expanding Boiler Upgrade Scheme commercial pilots, tightening F-Gas phase-down eliminating high-GWP refrigerants, and Net Zero commitments create urgent demand for compliant, efficient commercial heat pumps in the 20–50kW range currently underserved by R290 products.

Small commercial building owners and operators (offices, retail, schools, light industrial); M&E contractors and heat pump installers; local authorities and housing associations with small commercial assets; property developers targeting EPC-B/A ratings

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Vaillant Group	
Product	aroTHERM plus VWL 285/5 AS (commercial variant, ~28kW)
Pricing	Estimated £6,500–8,500 for 28kW unit ex-installer (list price not publicly quoted for commercial; trade pricing approx. £5,800–7,200)
Tech Specs	28.3kW heating capacity at A7/W35, COP 5.0 at A7/W35 / COP 3.4 at A2/W35, R290 refrigerant, split system, hydronic indoor unit, operating range -20°C to +43°C ambient, sound power 65 dB(A), SCOP 4.2 (medium climate)
STRENGTHS Established brand with strong UK installer network; MCS certified; R290 already in portfolio; excellent seasonal performance; comprehensive controls integration; pan-European service network WEAKNESSES Primarily residential-focused product stretched to small commercial; limited commercial-specific features (buffer vessel sizing, BMS integration, commercial warranty); higher price point; lead times can be long; indoor unit not designed for plantroom installation	
DIFFERENTIATION ANGLE Our product targets the commercial segment natively with plantroom-ready indoor hydraulic module, commercial BMS (BACnet/Modbus) as standard, 3-year commercial warranty, and a lower installed cost. We match R290 credentials while offering purpose-built commercial features Vaillant's residential product lacks.	

Clivet SpA

Product	WSAN-XEE2 30 (30kW air-to-water heat pump)
Pricing	Approximately £7,000–9,500 ex-works for 30kW class unit; UK distribution adds 15–25%
Tech Specs	30kW nominal heating, COP 3.8 at A7/W35, R410A refrigerant (transitioning range), hydronic split configuration, scroll compressor, EC fan motors, integrated hydraulic module option, BACnet/Modbus BMS interface, operating range -20°C to +46°C

STRENGTHS

Purpose-built commercial product with full BMS integration; strong European commercial reference base; robust hydraulic module; wide operating range; professional technical support

WEAKNESSES

Currently R410A (high GWP, facing phase-out); R290 version not yet in 30kW commercial class; Italian manufacturing with longer UK lead times; less established UK brand vs Vaillant; installer training less widespread in UK

DIFFERENTIATION ANGLE

Our R290 product directly addresses Clivet's refrigerant vulnerability; we offer future-proof compliance with F-Gas trajectory while matching their commercial feature set at a lower target price point of £4,500 vs their £7,000+ positioning.

Mitsubishi Electric

Product	Ecodan CAHV-R450YA-HPB (Hydrobox commercial, 40–50kW class)
Pricing	£10,000–14,000 for 45kW system (outdoor + indoor); smaller 25–30kW configurations £7,500–10,000 estimated
Tech Specs	45kW maximum heating output, R32 refrigerant (GWP 675), COP 3.94 at A7/W35, commercial hydrobox indoor unit with integrated controls, modular outdoor units, BACnet/Modbus standard, MCS certified, operating to -28°C ambient

STRENGTHS

Market-leading brand recognition; very strong UK installer and service network; MCS certified; proven commercial reliability; excellent cold-climate performance; comprehensive controls and monitoring platform (MELCloud)

WEAKNESSES

R32 refrigerant (GWP 675) faces future F-Gas restrictions; significantly higher price than target; minimum commercial unit sizes may be oversized for 30kW applications; product complexity may deter smaller installers; not R290

DIFFERENTIATION ANGLE

We undercut Mitsubishi Electric on price by 40–50% while offering superior refrigerant credentials (R290 GWP=3 vs R32 GWP=675), giving specifiers and building owners a future-proof, lower-capital solution for the 25–35kW commercial segment that Mitsubishi currently overshoots with larger systems.

Midea / Carrier (OEM)

Product	Midea M-Thermal Arctic 30kW (and Carrier AquaSnap re-brands)
Pricing	£3,200–4,500 ex-factory Asia; UK landed and distributed approximately £4,800–6,000 trade price
Tech Specs	30kW heating at A7/W35, COP 4.1 at A7/W35 / estimated COP 3.5–3.6 at A2/W35, R32 refrigerant, split hydronic system, DC inverter scroll compressor, Wi-Fi and BMS ready, EN 14511 tested, MCS application in progress for some models

STRENGTHS

Very competitive ex-factory cost enabling aggressive market pricing; good baseline efficiency; inverter compressor; improving quality and European certification progress; Carrier brand adds credibility in commercial sector

WEAKNESSES

R32 refrigerant sustainability concerns; MCS certification incomplete or recent for commercial range; UK service and spares infrastructure less mature; perception gap vs European brands in commercial specifications; BMS integration less proven in UK projects

DIFFERENTIATION ANGLE

Our product matches Midea's price competitiveness (target £4,500 vs their £4,800–6,000 UK trade) while offering R290 future-proofing, full MCS certification from launch, and a purpose-designed commercial indoor unit — addressing the two key weaknesses (refrigerant and certification) that currently limit Midea's commercial penetration.

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	500
Target Cost Ceiling	£4,500.00
Max Mass	Not computed

Research Sources

Title	Type	Year	Relevance
MCS 007 Heat Pump Products Standard	regulatory_standard	2026	Defines mandatory performance testing, documentation, and quality requirements for heat pumps sold under MCS certification in the UK market
EN 14511:2023 Air Conditioners, Liquid Chilling Packages and Heat Pumps	technical_standard	2026	Primary performance test standard referenced by MCS for rating COP at A2/W35 and other conditions; directly governs product performance claims
UK F-Gas Regulation (SI 2015/310 as retained and amended)	regulatory_standard	2026	Governs refrigerant use, phase-down schedules, and handling requirements; R290 exemption from quota system is a key commercial advantage
EN 378:2021 Refrigerating Systems and Heat Pumps – Safety Requirements	technical_standard	2026	Safety standard for flammable refrigerant (A3 class) charge limits, installation requirements, and risk assessment applicable to R290 systems
BEIS Heat and Buildings Strategy 2021	government_policy	2026	Sets out UK roadmap for heat pump deployment, commercial incentives, and phasing out fossil fuel heating in commercial buildings, defining market context

Commission Regulation (EU) 813/2013 – Ecodesign for Space Heaters (UK retained)	regulatory_standard	2026	Mandatory seasonal space heating energy efficiency (s) and labelling requirements applicable to heat pumps placed on UK market
EHPA European Heat Pump Market and Statistics Report 2023	market_research	2026	Provides market sizing, growth trajectories, and technology share data for commercial heat pump segment across European markets including UK
UK Pressure Equipment (Safety) Regulations 2016 (SI 2016/1105)	regulatory_standard	2026	Governs design, manufacture, and conformity assessment of pressure-bearing components including refrigerant circuits and hydronic components above 0.5 bar

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2.1 MCS 007 [D]

MCS Heat Pump Products Standard

Status: not-started

Owner: Regulatory/Compliance Engineer + Product Manager

Summary

Mandatory UK certification scheme for heat pumps to be eligible for BUS grant funding and installer scheme membership; requires third-party performance testing, product documentation, and factory quality audits

Applicability

UK market; required for any heat pump sold through MCS-certified installers or claiming BUS eligibility

Engineering Impact

Performance must be tested at accredited laboratory at defined rating points including A2/W35; COP e3.5 must be demonstrated; product documentation pack and quality management system required

EVIDENCE REQUIRED

Third-party laboratory test reports to EN 14511, Declaration of Performance, technical data sheet, installation manual, quality management system audit certificate (ISO 9001 or equivalent)

GAP ACTION

Engage MCS-approved test body (e.g., Kiwa, BRE, Intertek) for pre-certification assessment; initiate QMS documentation; budget £25–40k for full certification process

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.2 EN 14511:2023 ^[D]

Air Conditioners, Liquid Chilling Packages and Heat Pumps for Space Heating and Cooling

Status: not-started

Owner: Thermal/Refrigeration Design Engineer

Summary

Specifies test conditions, methods, and calculation procedures for determining heating capacity and COP; mandatory reference standard for MCS and ErP compliance rating

Applicability

All heat pump performance claims and MCS/ErP compliance documentation; test points include A2/W35, A7/W35, A7/W45

Engineering Impact

Defines A2/W35 as the key rating condition; system design must achieve COP e3.5 at this point; evaporator defrost cycle losses must be included in steady-state and seasonal calculations

EVIDENCE REQUIRED

Accredited laboratory test report covering all required rating points with measurement uncertainty statements; calibration records for test equipment

GAP ACTION

Design refrigerant circuit and heat exchanger sizing to target COP e3.8 at A7/W35 to provide margin for A2/W35 e3.5; commission prototype testing at UKAS-accredited facility

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.3 EN 378:2021 ^[D]

Refrigerating Systems and Heat Pumps – Safety and Environmental Requirements

Status: not-started

Owner: Safety Engineer + Refrigeration Design Engineer

Summary

Four-part standard covering safety requirements for design, construction, testing, installation, and operation of refrigerating systems; critical for A3 (flammable) refrigerant R290

Applicability

Applies to design of refrigerant circuit; particularly Parts 1 and 2 for R290 charge limits, leak detection requirements, and installation in or near buildings

Engineering Impact

R290 charge must be minimised; hydronic split architecture (refrigerant confined to outdoor unit) strongly preferred; outdoor unit must carry IP44 minimum; pressure relief and charge labelling mandatory; risk assessment required

EVIDENCE REQUIRED

Risk assessment documentation per EN 378-1; pressure test certificates; conformity declaration referencing EN 378; installation manual covering safe handling of R290

GAP ACTION

Conduct EN 378-1 risk assessment at concept stage; design outdoor unit to contain entire R290 circuit with maximum charge <1.5 kg; engage notified body for conformity assessment review

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.4 SI 2016/1105 [D]

UK Pressure Equipment (Safety) Regulations 2016

Status: not-started

Owner: Mechanical/Pressure Systems Engineer

Summary

Implements PED requirements in UK law; governs design, manufacture, and conformity assessment of pressure equipment and assemblies; applies to refrigerant circuit and hydronic circuit components

Applicability

Refrigerant-side components (compressor, heat exchangers, valves, pipework); hydronic circuit components above 0.5 bar working pressure

Engineering Impact

Pressure vessels and piping assemblies must be designed and tested to PSSR; material selection, welding procedures, and pressure testing protocols must be documented; CE/UKCA marking of pressure assemblies required

EVIDENCE REQUIRED

Technical file including pressure calculations, material certificates, welding procedure qualifications, hydrostatic or pneumatic test records, conformity assessment by approved body for Category II+ equipment

GAP ACTION

Map all pressure-bearing components to PSSR categories; engage UK Approved Body for Category II assessment; ensure brazed plate heat exchanger suppliers provide PSSR-compliant documentation

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

2.5 EU 813/2013 (UK retained) [D]

Ecodesign Regulation for Space Heaters and Combination Heaters

Status: not-started

Owner: Compliance/Regulatory Engineer

Summary

Sets mandatory minimum seasonal space heating energy efficiency (s) thresholds and information requirements for heat pumps; requires ErP label and product fiche; minimum s e125% for heat pumps

Applicability

All space heating products placed on UK market including air-source heat pumps up to 400kW; product fiche and label must accompany product

Engineering Impact

Seasonal efficiency calculation per EN 14825 required across heating season bin temperatures; s must be calculated and declared; noise levels must be tested and declared

EVIDENCE REQUIRED

EN 14825 seasonal performance calculation report; sound power level test report per EN 12102; product fiche completed per Annex IV of Regulation 813/2013; technical documentation file

GAP ACTION

Commission EN 14825 bin calculation alongside EN 14511 testing; prepare product fiche template; register product on EPREL database (UK equivalent via OPSS); ensure ErP label is print-ready before market launch

Data Sources

[Regulatory LLM] Standards extracted from context

Overall Section Grade: D

3. Sizing & Spatial Optimisation [D]

INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

System Envelope

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Internal Floor Area	25 m²
Internal Volume	75 m³
Rules Domain	hvac_heat_pumps

Module Allocation Zone Table

Module	L × W × H (mm)	Area (m²)	Mount
compressor_module	2236 × 2236 × 2400	5	floor
evaporator_coil_module	2236 × 2236 × 2400	5	floor
air_handling_module	2236 × 2236 × 2400	5	floor
refrigerant_routing_module	2236 × 2236 × 2400	5	floor
condenser_module	2236 × 2236 × 2400	5	floor
power_electronics_module	2236 × 2236 × 2400	5	floor
system_controller_module	2236 × 2236 × 2400	5	floor
chassis_and_enclosure	2236 × 2236 × 2400	5	floor
indoor_hydraulic_module	2236 × 2236 × 2400	5	floor

Conflicts

- Total module footprint (45.0m2) exceeds envelope budget (21.3m2)



Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

Overall Section Grade: B

4.1 R290 Scroll Compressor Assembly [D]

Maturity: draft

Purpose

Compresses low-pressure R290 vapor to high-pressure, high-temperature gas, driving the thermodynamic cycle.

Why It Matters

It dictates the overall capacity (30kW target), electrical efficiency (COP e 3.5 at A2/W35), and acoustic footprint of the system, acting as the most expensive component strictly governed by ATEX/EN 378 safety regimes.

Technical Description

The compressor assembly is the prime mover of the 30kW hydronic split system, engineered specifically for A3 flammable refrigerants (R290). Propane has excellent thermodynamic properties, allowing for higher discharge temperatures and exceptional seasonal efficiencies (SCOP > 3.2), but its high solubility in oil requires specific PoE or PAG lubricants to prevent viscosity breakdown. The selected inverter-driven scroll compressor provides precise capacity modulation to match the varying heat loads of small commercial buildings, dynamically varying speed to maintain optimal operation.

To meet EN 378 safety specifications, the compressor wiring and terminals must feature sealed, intrinsically safe connections to prevent ignition in the event of an R290 leak. Because commercial environments (like offices and schools) are sensitive to noise, the module is mechanically decoupled from the chassis using tuned elastomeric anti-vibration mounts and wrapped in a bespoke acoustic jacket. Continuous heating via a crankcase heater prevents liquid-refrigerant migration into the oil during off-cycles, ensuring safe, lubricated start-ups even at -20°C ambient temperatures.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.2 Air-to-Refrigerant Evaporator Coil [D]

Maturity: draft

Purpose

Extracts thermal energy from outdoor ambient air to boil liquid R290 into a gas.

Why It Matters

Coil efficiency directly defines the heat pump's seasonal performance (SCOP), while its internal volume dictates the total R290 charge, driving regulatory compliance complexity.

Technical Description

This highly engineered fin-and-tube heat exchanger acts as the evaporator during heating mode. It is scaled to extract sufficient ambient heat at the critical A2/W35 (2°C air) and A7/W35 rating points to achieve the mandatory COP of e3.5 and e3.8 respectively for MCS 007 certification. Advanced hydrophilic coatings on the louvered aluminum fins promote rapid condensate shedding, delaying the onset of frost bridging and extending the time between energy-intensive defrost cycles.

The geometry of the copper circuitry is precision-designed to minimize the internal volume, strictly limiting the maximum R290 charge to under 1.5 kg per circuit. This charge optimization is paramount for bypassing stricter EN 378 siting requirements for occupied spaces, while relying heavily on multi-port distributors to ensure uniform refrigerant allocation across the coil face. During sub-zero operation, ice buildup is inevitable, so the module integrates seamlessly with a base pan heater and drainage tray to securely channel meltwater away from the chassis without re-freezing.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.3 Active Airflow & Fan Assembly [D]

Maturity: draft

Purpose

Induces optimal ambient air mass flow across the evaporator coil to sustain evaporative heat transfer.

Why It Matters

It establishes the air-side heat transfer rate critical for hitting COP targets, while its acoustic performance sets the capability of the product to meet stringent town-planning noise limits, a major barrier in UK building retrofits.

Technical Description

To handle the massive airflow demands of a 30kW unit while remaining whisper-quiet, the active airflow module utilizes dual large-diameter Electronically Commutated (EC) axial fans. These highly efficient fans comply strictly with the Ecodesign (ErP) Directive EU 813/2013 by varying their speed dynamically based on the evaporation pressure. The bionic, swept-blade profile of the impellers drastically reduces trailing-edge vortex shedding, minimizing broadband acoustic noise and helping the unit comply with strict local planning authority noise limits for commercial buildings near residential zones.

The fans are mounted into geometrically optimized venturi shrouds integrated into the unit's exterior enclosure, preventing backward turbulence and ensuring air is pulled evenly across the entirety of the evaporator face. In combination with tachometer feedback, the control module can instantly detect blockages (like winter snowdrifts or autumn debris) by measuring deviations in fan load, prompting warning alerts through the BMS interface.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.4 Refrigerant Control & Trapeze Sub-assembly

[D]

Maturity: draft

Purpose

Meters refrigerant flow and reversing cycle direction for heating/defrost state changes.

Why It Matters

Controls the thermodynamic state and ensures physical pressure containment; precise EEV control separates basic systems from high-COP (3.8+) Ecodesign-compliant systems.

Technical Description

This intricate network of brazed copper pipework and electromechanical valves is the central nervous system of the refrigeration cycle. At its core is the Electronic Expansion Valve (EEV), which dynamically steps down the high-pressure liquid R290 to a low-pressure two-phase mixture using a high-resolution stepper motor. This precise metering ensures the returning gas maintains the optimal superheat, maximizing efficiency and protecting the compressor from liquid slugging. Because R290 operates at moderately high pressures, all piping and valves strictly fall under the UK Pressure Equipment (Safety) Regulations (SI 2016/1105) Category II, mandating certified non-destructive testing of brazed joints.

The 4-way reversing valve is critical for winter reliability. When frost accumulates on the evaporator, the 4-way valve flips the flow of the entire cycle, temporarily turning the outdoor coil into a condenser to rapidly melt the ice using building heat. Dual analogue pressure transducers monitor the high and low lines for optimal control, while redundant mechanical safety pressure switches are hardwired inline to mechanically break the compressor circuit, mitigating over-pressurization regardless of software state.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.5 Brazed Plate Heat Exchanger (Condenser) [D]

Maturity: draft

Purpose

Transfers heat from the high-temperature R290 refrigerant to the building's hydronic heating circuit.

Why It Matters

It is the primary thermal bridge conveying generated heat to the building; its efficiency directly dictates system COP, whilst allowing R290 to remain outside the building envelope.

Technical Description

Positioned securely within the outdoor unit to maintain the strict hydronic split architecture, the Condenser Module isolates the A3 flammable R290 entirely to the exterior of the commercial premises. The core of this module is an asymmetrical Stainless Steel Brazed Plate Heat Exchanger (BPHE). Counter-flow geometry enables highly efficient heat transfer, cooling the superheated R290 vapor down into a subcooled liquid while simultaneously heating the secondary water circuit to target flow temperatures (typically 35°C to 55°C).

Because of the aggressive heat gradients and repetitive thermal expansion experienced during defrost cycling, the BPHE is subject to intense thermal fatigue. It is mounted vertically on isolated rubber damper pads and wrapped entirely in closed-cell elastomeric foam to prevent heat loss and avoid external condensation. The module directly impacts EN 14511 performance; minimizing the R290-side internal volume here drastically reduces total refrigerant system charge, solving one of the main compliance barriers associated with natural refrigerant systems.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.6 Mains & Inverter Drive Enclosure [D]

Maturity: draft

Purpose

Conditions mains AC power and drives the variable-speed compressor and fans with high power-factor efficiency.

Why It Matters

It drives unit reliability and prevents electrical grid pollution, while its physical isolation from the R290 circuit is fundamental to EN 378 / ATEX fire safety.

Technical Description

Modern high-efficiency heat pumps rely heavily on variable capacity adjustment, requiring complex, robust power electronics. This module ingests 400V 3-phase commercial mains power, filtering it through aggressive EMI/RFI stages to comply with UKCA Electromagnetic Compatibility (EMC) standards. Active Power Factor Correction (PFC) circuitry conditions the power before rectifying it to a high-voltage DC link. High-power IGBTs (Insulated-Gate Bipolar Transistors) then invert this energy to provide perfectly modulated variable-frequency signals to the compressor and fan motors.

Due to the presence of R290, this electrical box must be isolated from the refrigeration circuit. It is housed in a fireproof, IP54+ sealed metallic sub-enclosure. Because the IGBTs generate substantial waste heat, an external aluminum heatsink protrudes into the outdoor unit's main fan airstream to provide continuous active cooling. This thermal management ensures the electronics do not derate heavily during extreme summer operations (e.g., domestic hot water charging at +35°C ambient).

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.7 Main Controls & BMS Interface [D]

Maturity: draft

Purpose

Executes the system state machine, optimizes efficiency loops, and interfaces with external building networks.

Why It Matters

Software algorithms here directly distinguish underperforming hardware assemblies from premium-tier, high-SCOP systems, while the BMS interface captures the commercial specifier market.

Technical Description

The logic and brain center of the 30kW commercial heat pump, the System Controller Module manages the complex thermodynamic algorithms required for seasonal capability and compressor envelope protection. By constantly reading ambient, discharge, evaporating, and water return temperatures, the software determines the precise superheat requirement, dynamically sending steps to the EEV and adjusting the compressor RPM. Within its software stack reside the strict anti-legionella cycles, weather compensation curves, and autonomous defrost heuristics necessary to maximize EN 14825 seasonal efficiency scores.

Crucially for the UK small commercial market, this controller features native commercial Building Management System (BMS) integrations, avoiding the need for expensive third-party wrappers. Modbus and BACnet protocols allow integration directly into plantroom aggregators for facilities managers, providing a strong point of differentiation over residential-skewed products. It also logs errors, timestamps run-hours, and manages all required MCS fault-reporting logic compliant with grid connection codes.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.8 Structural Frame & Enclosure [D]

Maturity: draft

Purpose

Shields internal components from environmental degradation, attenuates sound, and physically defines the product footprint.

Why It Matters

It establishes the first impression of product quality, guarantees long-term survival in aggressive environments, and provides necessary structural integrity and acoustic limits.

Technical Description

Built for outdoor roof-mount or ground-level installations, the enclosure must resist harsh UK weather conditions for a 15+ year commercial lifespan. Manufactured from heavy-gauge galvanized steel and finished with a 500-hour salt-spray-resistant polyester powder coat, the modular chassis prevents structural corrosion. Crucially, as the refrigerant is R290, EN 378 requires the outdoor box construction to safely disperse any leaked internal gas preventing explosive pooling; this mandates deliberate ventilation gaps or ATEX-rated extraction pathways at heavy gas accumulation zones at the base pan.

The chassis design serves a dense structural purpose—incorporating dedicated forklift channels and lifting eyelets to facilitate crane-lifts to commercial building rooftops. Internally, thick layers of hydrophobic acoustic foam mute the mechanical rumble of the compressor and flow whine of the EEV, enabling the unit to meet the sub-65 dB(A) commercial sound power requirements essential for UK planning permission approvals. Easy-access service doors are engineered into the exterior layout for immediate F-gas checks and preventative maintenance.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.9 Indoor Hydronic Control Station [D]

Maturity: draft

Purpose

Circulates the heated water through the building's secondary circuit and provides necessary hydronic safety devices.

Why It Matters

It completes the product offering as a system, translating the ASHP's raw heat generation into manageable, safe, and distributed hot water for immediate plantroom integration.

Technical Description

Crucial to the 'hydronic split' architecture, this module resides entirely inside the commercial building (usually wall-mounted in the plantroom). Because all R290 refrigerant is sealed securely in the outdoor unit, installing this indoor module involves only standard plumbing, neutralizing any EN 378 refrigerant-volume constraints inside the premises. It functions as the motive and safety heart of the physical water circuit.

Inside the painted steel casing is a commercial-grade, high-head circulation pump capable of overcoming the pressure drops of multi-story office layouts and pushing steady flow to buffer vessels. Designed to comply with Pressure Equipment Safety Regulations (SI 2016/1105), it integrates an expansion vessel to absorb volumetric water expansion during heating, a Safety Pressure Relief Valve (PRV) to mitigate closed-loop overpressure, and a magnetic filter to capture legacy black-iron pipe sludge from old gas-boiler retrofits. Integrated flow meters inform the outdoor controller, confirming active water flow before allowing the compressor to strike.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
R290 Scroll Compressor Assembly	Refrigerant ignition at terminal block	Not computed	Not computed	5×1=5	Use ATEX certified hermetic terminal seals and ATEX EX-d enclosure design.
R290 Scroll Compressor Assembly	Liquid slugging at low ambient temps (-20°C)	Not computed	Not computed	4×2=8	Implement strict superheat control algorithms and install an oversized suction accumulator.
R290 Scroll Compressor Assembly	Bearing seizure from oil wash-out	Not computed	Not computed	4×2=8	Ensure crankcase heater is interlocked with control board to operate continuously when compressor is off.
Air-to-Refrigerant Evaporator Coil	Loss of R290 containment via vibration fatigue	Not computed	Not computed	4×2=8	Finite Element Analysis (FEA) on tube sheet; optimize tube expansion constraints.
Air-to-Refrigerant Evaporator Coil	Defrost pan ice accumulation	Not computed	Not computed	3×3=9	Integrate redundantly rated self-regulating trace heating cables in the base pan.
Air-to-Refrigerant Evaporator Coil	Inadequate capacity at -15°C	Not computed	Not computed	3×2=6	Oversize fin density slightly while checking air-side pressure drop; model performance envelope heavily.
Active Airflow & Fan Assembly	Blade fracture from ice strike	Not computed	Not computed	4×2=8	Design shroud and coil distance to maintain >50mm gap; standardise reinforced polyamide/fiberglass composite blades.

Active Airflow & Fan Assembly	Non-compliant ErP acoustic output	Not computed	Not computed	3×2=6	Test extensively in hemi-anechoic chamber per EN 12102 early in prototype phase.
Active Airflow & Fan Assembly	Motor overheat due to restricted flow	Not computed	Not computed	3×3=9	Include built-in thermal overload protection in EC motor internal logic combined with fault-reporting.
Refrigerant Control & Trapeze Sub-assembly	R290 leak via brazing defect	Not computed	Not computed	5×2=10	Implement nitrogen-purged automated induction brazing; 100% helium leak test post-assembly.
Refrigerant Control & Trapeze Sub-assembly	Incomplete valve shifting causing bypass	Not computed	Not computed	3×3=9	Ensure 4-way valve is sized correctly for minimum pressure differential needed to throw internal slide.
Refrigerant Control & Trapeze Sub-assembly	Pressure Equipment Directive (PED) non-compliance	Not computed	Not computed	4×1=4	Maintain robust technical files, material certs, and engage UK Approved Body early for Category II assessment.
Brazed Plate Heat Exchanger (Condenser)	BPHE freezing due to low flow	Not computed	Not computed	5×2=10	Mandate high-precision frost protection algorithm utilizing both physical flow switches and redundant temp sensors. Mandate anti-freeze in commercial loops.
Brazed Plate Heat Exchanger (Condenser)	Refrigerant migration into water	Not computed	Not computed	4×1=4	Specify BPHE to resist >45 bar test pressure; mandate safety relief valve (PRV) on hydronic circuit indoor module.
Brazed Plate Heat Exchanger (Condenser)	Heat loss through casing	Not computed	Not computed	2×3=6	Ensure vapor-tight seal utilizing bespoke molded EPP/elastomeric foam enclosure.
Mains & Inverter Drive Enclosure	R290 infiltration into electrical box	Not computed	Not computed	5×1=5	Design isolated compartment with forced ventilation or guaranteed IP65 sealed cable glands complying with EN 60079-15.

Mains & Inverter Drive Enclosure	EMC emissions failing UKCA tests	Not computed	Not computed	4×2=8	Integration of commercial-grade multi-stage line filters; perform pre-compliance testing early.
Mains & Inverter Drive Enclosure	Heatsink airflow restriction causing derating	Not computed	Not computed	3×3=9	Position massive finned heatsink directly in the exhaust flow cavity of the main axial fans.
Main Controls & BMS Interface	Defrost algorithm logic failure	Not computed	Not computed	4×2=8	Employ redundant temperature thresholds for initiating defrost; feature factory override capabilities.
Main Controls & BMS Interface	Loss of grid sync or power spike damaging MCU	Not computed	Not computed	3×2=6	Opto-isolate digital inputs and provide robust secondary surge protection on main 24V bus.
Main Controls & BMS Interface	Installer wiring error on interface	Not computed	Not computed	2×4=8	Design intuitively segregated, color-coded, screwless WAGO-style terminal blocks for external hookups.
Structural Frame & Enclosure	R290 pooling in lower chassis	Not computed	Not computed	5×2=10	Design chassis with specialized louvers and base pan openings to allow heavy propane to drain downward and dissipate naturally.
Structural Frame & Enclosure	Corrosion in coastal environments	Not computed	Not computed	3×3=9	Require electro-galvanization prior to heavy powdercoat; mandate C4 corrosivity minimum specs.
Structural Frame & Enclosure	Access panel rattling due to vibration	Not computed	Not computed	2×4=8	Integrate EPDM continuous rubber gaskets on all removable sheet metal panels.
Indoor Hydronic Control Station	Loss of flow causing outdoor BPHE freeze	Not computed	Not computed	5×2=10	Dual verification of flow by combining physical paddle switch with electronic delta-T checks via NTC thermistors.

Indoor Hydronic Control Station	Sludge ingress from legacy pipework	Not computed	Not computed	3×4=12	Ship module with heavy-duty commercial cyclone/magnetic separator to shield the outdoor BPHE.
Indoor Hydronic Control Station	Vessel sizing mismatch for building	Not computed	Not computed	2×3=6	Include an externally rated expansion port, advising installers that internal volume buffers minimums, not maximums.

Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	9 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success